

SUMMARY

One of the exciting developments in finance over the last 40 years has been the growth of derivatives markets. In many situations, both hedgers and speculators find it more attractive to trade a derivative on an asset than to trade the asset itself. Some derivatives are traded on exchanges; others are traded by financial institutions, fund managers, and corporations in the over-the-counter market, or added to new issues of debt and equity securities. Much of this book is concerned with the valuation of derivatives. The aim is to present a unifying framework within which all derivatives—not just options or futures—can be valued.

In this chapter we have taken a first look at forward, futures, and option contracts. A forward or futures contract involves an obligation to buy or sell an asset at a certain time in the future for a certain price. There are two types of options: calls and puts. A call option gives the holder the right to buy an asset by a certain date for a certain price. A put option gives the holder the right to sell an asset by a certain date for a certain price. Forwards, futures, and options trade on a wide range of different underlying assets.

The success of derivatives can be attributed to their versatility. They can be used by: hedgers, speculators, and arbitrageurs. Hedgers are in the position where they face risk associated with the price of an asset. They use derivatives to reduce or eliminate this risk. Speculators wish to bet on future movements in the price of an asset. They use derivatives to get extra leverage. Arbitrageurs are in business to take advantage of a discrepancy between prices in two different markets. If, for example, they see the futures price of an asset getting out of line with the cash price, they will take offsetting positions in the two markets to lock in a profit.

FURTHER READING

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Miller, M. H. “Financial Innovation: Achievements and Prospects,” *Journal of Applied Corporate Finance*, 4 (Winter 1992): 4–11.

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Short Concept Questions

- 1.1. What is a derivative?
- 1.2. Explain the two main ways that derivatives trade?
- 1.3. What are the changes that have taken place in the regulation of the OTC market since the 2008 financial crisis?
- 1.4. Which is bigger, the exchange-traded market or the OTC market?

- 1.5. What is the difference between a long forward position and a short forward position?
- 1.6. How do (a) forward and (b) futures contracts trade?
- 1.7. What is the difference between a forward contract to buy an asset at \$30 and a call option to buy the same asset for \$30?
- 1.8. Explain carefully the difference between hedging, speculation, and arbitrage.
- 1.9. Explain the meaning of bid and ask quotes.
- 1.10. What gave rise to Jerome Kerviel's big loss?

Practice Questions

- 1.11. Explain carefully the difference between selling a call option and buying a put option.
- 1.12. An investor enters into a short forward contract to sell 100,000 British pounds for U.S. dollars at an exchange rate of 1.3000 USD per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is (a) 1.2900 and (b) 1.3200?
- 1.13. A trader enters into a short cotton futures contract when the futures price is 50 cents per pound. The contract is for the delivery of 50,000 pounds. How much does the trader gain or lose if the cotton price at the end of the contract is (a) 48.20 cents per pound and (b) 51.30 cents per pound?
- 1.14. Suppose that you write a put contract with a strike price of \$40 and an expiration date in 3 months. The current stock price is \$41 and the contract is on 100 shares. What have you committed yourself to? How much could you gain or lose?
- 1.15. You would like to speculate on a rise in the price of a certain stock. The current stock price is \$29 and a 3-month call with a strike price of \$30 costs \$2.90. You have \$5,800 to invest. Identify two alternative investment strategies, one in the stock and the other in an option on the stock. What are the potential gains and losses from each?
- 1.16. Suppose that you own 5,000 shares worth \$25 each. How can put options be used to provide you with insurance against a decline in the value of your holding over the next 4 months?
- 1.17. When first issued, a stock provides funds for a company. Is the same true of a stock option? Discuss.
- 1.18. Explain why a futures contract can be used for either speculation or hedging.
- 1.19. Suppose that a March call option to buy a share for \$50 costs \$2.50 and is held until March. Under what circumstances will the holder of the option make a profit? Under what circumstances will the option be exercised? Draw a diagram illustrating how the profit from a long position in the option depends on the stock price at maturity of the option.
- 1.20. Suppose that a June put option to sell a share for \$60 costs \$4 and is held until June. Under what circumstances will the seller of the option (i.e., the party with the short position) make a profit? Under what circumstances will the option be exercised? Draw a diagram illustrating how the profit from a short position in the option depends on the stock price at maturity of the option.
- 1.21. It is May and a trader writes a September call option with a strike price of \$20. The stock price is \$18 and the option price is \$2. Describe the trader's cash flows if the option is held until September and the stock price is \$25 at that time.

- 1.22. A trader writes a December put option with a strike price of \$30. The price of the option is \$4. Under what circumstances does the trader make a gain?
- 1.23. A company knows that it is due to receive a certain amount of a foreign currency in 4 months. What type of option contract is appropriate for hedging?
- 1.24. A U.S. company expects to have to pay 1 million Canadian dollars in 6 months. Explain how the exchange rate risk can be hedged using (a) a forward contract and (b) an option.
- 1.25. A trader enters into a short forward contract on 100 million yen. The forward exchange rate is \$0.0090 per yen. How much does the trader gain or lose if the exchange rate at the end of the contract is (a) \$0.0084 per yen and (b) \$0.0101 per yen?
- 1.26. The CME Group offers a futures contract on long-term Treasury bonds. Characterize the traders likely to use this contract.
- 1.27. “Options and futures are zero-sum games.” What do you think is meant by this?
- 1.28. Describe the profit from the following portfolio: a long forward contract on an asset and a long European put option on the asset with the same maturity as the forward contract and a strike price that is equal to the forward price of the asset at the time the portfolio is set up.
- 1.29. In the 1980s, Bankers Trust developed *index currency option notes* (ICONS). These were bonds in which the amount received by the holder at maturity varied with a foreign exchange rate. One example was its trade with the Long Term Credit Bank of Japan. The ICON specified that if the yen/USD exchange rate, S_T , is greater than 169 yen per dollar at maturity (in 1995), the holder of the bond receives \$1,000. If it is less than 169 yen per dollar, the amount received by the holder of the bond is

$$1,000 - \max\left[0, 1,000\left(\frac{169}{S_T} - 1\right)\right]$$

When the exchange rate is below 84.5, nothing is received by the holder at maturity. Show that this ICON is a combination of a regular bond and two options.

- 1.30. On July 1, 2021, a company enters into a forward contract to buy 10 million Japanese yen on January 1, 2022. On September 1, 2021, it enters into a forward contract to sell 10 million Japanese yen on January 1, 2022. Describe the payoff from this strategy.
- 1.31. Suppose that USD/sterling spot and forward exchange rates are as follows:

Spot	1.2580
90-day forward	1.2556
180-day forward	1.2518

What opportunities are open to an arbitrageur in the following situations?

- (a) A 180-day European call option to buy £1 for \$1.22 costs 2 cents.
- (b) A 90-day European put option to sell £1 for \$1.29 costs 2 cents.
- 1.32. A trader buys a call option with a strike price of \$30 for \$3. Does the trader ever exercise the option and lose money on the trade? Explain your answer.
- 1.33. A trader sells a put option with a strike price of \$40 for \$5. What is the trader’s maximum gain and maximum loss? How does your answer change if it is a call option?
- 1.34. “Buying a put option on a stock when the stock is owned is a form of insurance.” Explain this statement.

- 1.35. On May 21, 2020, as indicated in Table 1.2, the spot ask price of Apple stock is \$316.50 and the ask price of a call option with a strike price of \$320 and a maturity date of September is \$21.70. A trader is considering two alternatives: buy 100 shares of the stock and buy 100 September call options. For each alternative, what is (a) the upfront cost, (b) the total gain if the stock price in September is \$400, and (c) the total loss if the stock price in September is \$300. Assume that the option is not exercised before September and positions are unwound at option maturity.
- 1.36. What is arbitrage? Explain the arbitrage opportunity when the price of a dually listed mining company stock is \$50 (USD) on the New York Stock Exchange and \$60 (CAD) on the Toronto Stock Exchange. Assume that the exchange rate is such that 1 U.S. dollar equals 1.21 Canadian dollars. Explain what is likely to happen to prices as traders take advantage of this opportunity.
- 1.37. Trader A enters into a forward contract to buy an asset for \$1,000 in one year. Trader B buys a call option to buy the asset for \$1,000 in one year. The cost of the option is \$100. What is the difference between the positions of the traders? Show the profit as a function of the price of the asset in one year for the two traders.
- 1.38. In March, a U.S. investor instructs a broker to sell one July put option contract on a stock. The stock price is \$42 and the strike price is \$40. The option price is \$3. Explain what the investor has agreed to. Under what circumstances will the trade prove to be profitable? What are the risks?
- 1.39. A U.S. company knows it will have to pay 3 million euros in three months. The current exchange rate is 1.1500 dollars per euro. Discuss how forward and options contracts can be used by the company to hedge its exposure.
- 1.40. A stock price is \$29. A trader buys one call option contract on the stock with a strike price of \$30 and sells a call option contract on the stock with a strike price of \$32.50. The market prices of the options are \$2.75 and \$1.50, respectively. The options have the same maturity date. Describe the trader's position.
- 1.41. The price of gold is currently \$1,200 per ounce. The forward price for delivery in 1 year is \$1,300 per ounce. An arbitrageur can borrow money at 3% per annum. What should the arbitrageur do? Assume that the cost of storing gold is zero and that gold provides no income.
- 1.42. On May 21, 2020, an investor owns 100 Apple shares. As indicated in Table 1.3, the share price is about \$316 and a December put option with a strike price of \$290 costs \$21.30. The investor is comparing two alternatives to limit downside risk. The first involves buying one December put option contract with a strike price of \$290. The second involves instructing a broker to sell the 100 shares as soon as Apple's price reaches \$290. Discuss the advantages and disadvantages of the two strategies.
- 1.43. A bond issued by Standard Oil some time ago worked as follows. The holder received no interest. At the bond's maturity the company promised to pay \$1,000 plus an additional amount based on the price of oil at that time. The additional amount was equal to the product of 170 and the excess (if any) of the price of a barrel of oil at maturity over \$25. The maximum additional amount paid was \$2,550 (which corresponds to a price of \$40 per barrel). Show that the bond is a combination of a regular bond, a long position in call options on oil with a strike price of \$25, and a short position in call options on oil with a strike price of \$40.

- 1.44. Suppose that in the situation of Table 1.1 a corporate treasurer said: “I will have £1 million to sell in 6 months. If the exchange rate is less than 1.19, I want you to give me 1.19. If it is greater than 1.25, I will accept 1.25. If the exchange rate is between 1.19 and 1.25, I will sell the sterling for the exchange rate.” How could you use options to satisfy the treasurer?

Short Concept Questions

- 2.1. Define the terms *open interest* and *trading volume*.
- 2.2. What is the difference between a *local* and a *futures commission merchant*.
- 2.3. What is the purpose of margin accounts?
- 2.4. What is the difference between a market order and a limit order?
- 2.5. What are the most important aspects of the design of a new futures contract?
- 2.6. How do the margin requirements for a small retail trader differ from those for a member of the exchange clearinghouse?
- 2.7. Explain the two ways derivatives trade in the OTC market?
- 2.8. Explain whether interest is paid on (a) initial margin and (b) variation margin in a futures trade.
- 2.9. Explain how hedge accounting works.
- 2.10. List the differences between forward and futures contracts.

Practice Questions

- 2.11. Suppose that you enter into a short futures contract to sell July silver for \$17.20 per ounce. The size of the contract is 5,000 ounces. The initial margin is \$4,000, and the maintenance margin is \$3,000. What change in the futures price will lead to a margin call? What happens if you do not meet the margin call?
- 2.12. Suppose that in September 2021 a company takes a long position in a contract on May 2022 crude oil futures. It closes out its position in March 2022. The futures price (per barrel) is \$48.30 when it enters into the contract, \$50.50 when it closes out its position, and \$49.10 at the end of December 2021. One contract is for the delivery of 1,000 barrels. What is the company's total profit? When is it realized? How is it taxed if it is (a) a hedger and (b) a speculator? Assume that the company has a December 31 year end.
- 2.13. What does a stop order to sell at \$2 mean? When might it be used? What does a limit order to sell at \$2 mean? When might it be used?
- 2.14. What differences exist in the way prices are quoted in the foreign exchange futures market, the foreign exchange spot market, and the foreign exchange forward market?
- 2.15. The party with a short position in a futures contract sometimes has options as to the precise asset that will be delivered, where delivery will take place, when delivery will take place, and so on. Do these options increase or decrease the futures price? Explain your reasoning.
- 2.16. Explain how margin accounts protect futures traders against the possibility of default.
- 2.17. A trader buys two July futures contracts on frozen orange juice concentrate. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is \$6,000 per contract, and the maintenance margin is \$4,500 per contract. What price change would lead to a margin call? Under what circumstances could \$2,000 be withdrawn from the margin account?

- 2.18. Show that, if the futures price of a commodity is greater than the spot price during the delivery period, then there is an arbitrage opportunity. Does an arbitrage opportunity exist if the futures price is less than the spot price? Explain your answer.
- 2.19. Explain the difference between a market-if-touched order and a stop order.
- 2.20. Explain what a stop-limit order to sell at 20.30 with a limit of 20.10 means.
- 2.21. At the end of one day a clearing house member is long 100 contracts, and the settlement price is \$50,000 per contract. The original margin is \$2,000 per contract. On the following day the member becomes responsible for clearing an additional 20 long contracts, entered into at a price of \$51,000 per contract. The settlement price at the end of this day is \$50,200. How much does the member have to add to its margin account with the exchange clearing house?
- 2.22. Explain why collateral requirements increased in the OTC market as a result of regulations introduced since the 2008 financial crisis.
- 2.23. The forward price of the Swiss franc for delivery in 45 days is quoted as 1.1000. The futures price for a contract that will be delivered in 45 days is 0.9000. Explain these two quotes. Which is more favorable for a trader wanting to sell Swiss francs?
- 2.24. Suppose you call your broker and issue instructions to sell one July hogs contract. Describe what happens.
- 2.25. "Speculation in futures markets is pure gambling. It is not in the public interest to allow speculators to trade on a futures exchange." Discuss this viewpoint.
- 2.26. What do you think would happen if an exchange started trading a contract in which the quality of the underlying asset was incompletely specified?
- 2.27. "When a futures contract is traded on the floor of the exchange, it may be the case that the open interest increases by one, stays the same, or decreases by one." Explain this statement.
- 2.28. Suppose that, on October 24, 2022, a company sells one April 2023 live cattle futures contract. It closes out its position on January 21, 2023. The futures price (per pound) is 121.20 cents when it enters into the contract, 118.30 cents when it closes out its position, and 118.80 cents at the end of December 2022. One contract is for the delivery of 40,000 pounds of cattle. What is the total profit? How is it taxed if the company is (a) a hedger and (b) a speculator? Assume that the company has a December 31 year end.
- 2.29. A cattle farmer expects to have 120,000 pounds of live cattle to sell in 3 months. The live cattle futures contract traded by the CME Group is for the delivery of 40,000 pounds of cattle. How can the farmer use the contract for hedging? From the farmer's viewpoint, what are the pros and cons of hedging?
- 2.30. It is July 2021. A mining company has just discovered a small deposit of gold. It will take 6 months to construct the mine. The gold will then be extracted on a more or less continuous basis for 1 year. Futures contracts on gold are available with delivery months every 2 months from August 2021 to December 2022. Each contract is for the delivery of 100 ounces. Discuss how the mining company might use futures markets for hedging.
- 2.31. Explain how CCPs work. What are the advantages to the financial system of requiring CCPs to be used for all standardized derivatives transactions between financial institutions?

- 2.32. Trader A enters into futures contracts to buy 1 million euros for 1.1 million dollars in three months. Trader B enters in a forward contract to do the same thing. The exchange rate (dollars per euro) declines sharply during the first two months and then increases for the third month to close at 1.1300. Ignoring daily settlement, what is the total profit of each trader? When the impact of daily settlement is taken into account, which trader has done better?
- 2.33. Explain what is meant by open interest. Why does the open interest usually decline during the month preceding the delivery month? On a particular day, there were 2,000 trades in a particular futures contract. This means that there were 2,000 buyers (going long) and 2,000 sellers (going short). Of the 2,000 buyers, 1,400 were closing out positions and 600 were entering into new positions. Of the 2,000 sellers, 1,200 were closing out positions and 800 were entering into new positions. What is the impact of the day's trading on open interest?
- 2.34. A company enters into a short futures contract to sell 5,000 bushels of wheat for 750 cents per bushel. The initial margin is \$3,000 and the maintenance margin is \$2,000. What price change would lead to a margin call? Under what circumstances could \$1,500 be withdrawn from the margin account?
- 2.35. Suppose that there are no storage costs for crude oil and the interest rate for borrowing or lending is 4% per annum. How could you make money if the June and December futures contracts for a particular year trade at \$50 and \$56, respectively?
- 2.36. What position is equivalent to a long forward contract to buy an asset at K on a certain date and a put option to sell it for K on that date.
- 2.37. A bank's derivatives transactions with a counterparty are worth +\$10 million to the bank and are cleared bilaterally. The counterparty has posted \$10 million of cash collateral. What credit exposure does the bank have?

The concept of cost of carry is sometimes useful. The cost of carry is the storage cost of the underlying asset plus the cost of financing it minus the income received from it. In the case of investment assets, the futures price is greater than the spot price by an amount reflecting the cost of carry. In the case of consumption assets, the futures price is greater than the spot price by an amount reflecting the cost of carry net of the convenience yield.

If we assume the capital asset pricing model is true, the relationship between the futures price and the expected future spot price depends on whether the return on the asset is positively or negatively correlated with the return on the stock market. Positive correlation will tend to lead to a futures price lower than the expected future spot price, whereas negative correlation will tend to lead to a futures price higher than the expected future spot price. Only when the correlation is zero will the theoretical futures price be equal to the expected future spot price.

FURTHER READING

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- Jarrow, R. A., and G. S. Oldfield. "Forward Contracts and Futures Contracts," *Journal of Financial Economics*, 9 (December 1981): 373–82.
- Richard, S., and S. Sundaresan. "A Continuous-Time Model of Forward and Futures Prices in a Multigood Economy," *Journal of Financial Economics*, 9 (December 1981): 347–72.
- Routledge, B. R., D. J. Seppi, and C. S. Spatt. "Equilibrium Forward Curves for Commodities," *Journal of Finance*, 55, 3 (2000) 1297–1338.

Short Concept Questions

- 5.1. What is the difference between an investment asset and a consumption asset?
- 5.2. Explain what happens when an investor shorts a certain stock.
- 5.3. What is the formula for the forward price of an investment asset that provides no income?
- 5.4. What is the formula for the forward price of an investment asset that provides a known dollar income?
- 5.5. What is the difference between a known yield and a known dollar income?
- 5.6. What is meant by index arbitrage?
- 5.7. What is the formula for the value of a forward contract in terms of the forward price?
- 5.8. What is the formula for a forward exchange rate in terms of the spot exchange rate?
- 5.9. Explain the meaning of *cost of carry* and *convenience yield*. What is the relationship between futures price, spot price, convenience yield, and cost of carry?
- 5.10. Under what circumstances is the futures price greater than the expected future spot price according to (a) Keynes and Hicks and (b) systematic risk considerations?

Practice Questions

- 5.11. What is the difference between the forward price and the value of a forward contract?
- 5.12. Suppose that you enter into a 6-month forward contract on a non-dividend-paying stock when the stock price is \$30 and the risk-free interest rate (with continuous compounding) is 5% per annum. What is the forward price?
- 5.13. A stock index currently stands at 350. The risk-free interest rate is 4% per annum (with continuous compounding) and the dividend yield on the index is 3% per annum. What should the futures price for a 4-month contract be?
- 5.14. Explain carefully why the futures price of gold can be calculated from its spot price and other observable variables whereas the futures price of copper cannot.
- 5.15. Explain why a foreign currency can be treated as an asset providing a known yield.
- 5.16. Is the futures price of a stock index greater than or less than the expected future value of the index? Explain your answer.
- 5.17. A 1-year long forward contract on a non-dividend-paying stock is entered into when the stock price is \$40 and the risk-free rate of interest is 5% per annum with continuous compounding.
 - (a) What are the forward price and the initial value of the forward contract?
 - (b) Six months later, the price of the stock is \$45 and the risk-free interest rate is still 5%. What are the forward price and the value of the forward contract?
- 5.18. The risk-free rate of interest is 7% per annum with continuous compounding, and the dividend yield on a stock index is 3.2% per annum. The current value of the index is 150. What is the 6-month futures price?
- 5.19. Assume that the risk-free interest rate is 4% per annum with continuous compounding and that the dividend yield on a stock index varies throughout the year. In February, May, August, and November, dividends are paid at a rate of 5% per annum. In other months, dividends are paid at a rate of 2% per annum. Suppose that the value of the index on July 31 is 1,300. What is the futures price for a contract deliverable in December 31 of the same year?
- 5.20. Suppose that the risk-free interest rate is 6% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?
- 5.21. Estimate the difference between short-term interest rates in Japan and the United States on May 21, 2020, from the information in Table 5.4.
- 5.22. The 2-month interest rates in Switzerland and the United States are, respectively, 1% and 2% per annum with continuous compounding. The spot price of the Swiss franc is \$1.0500. The futures price for a contract deliverable in 2 months is \$1.0500. What arbitrage opportunities does this create?
- 5.23. The spot price of silver is \$25 per ounce. The storage costs are \$0.24 per ounce per year payable quarterly in advance. Assuming that interest rates are 5% per annum for all maturities, calculate the futures price of silver for delivery in 9 months.
- 5.24. Suppose that F_1 and F_2 are two futures contracts on the same commodity with times to maturity, t_1 and t_2 , where $t_2 > t_1$. Prove that $F_2 \leq F_1 e^{r(t_2 - t_1)}$, where r is the interest rate

(assumed constant) and there are no storage costs. For the purposes of this problem, assume that a futures contract is the same as a forward contract.

- 5.25. When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the daily settlement process does leave the company exposed to some risk. Explain the nature of this risk. In particular, consider whether the company is better off using a futures contract or a forward contract when:
 - (a) The value of the foreign currency falls rapidly during the life of the contract.
 - (b) The value of the foreign currency rises rapidly during the life of the contract.
 - (c) The value of the foreign currency first rises and then falls back to its initial value.
 - (d) The value of the foreign currency first falls and then rises back to its initial value.
 Assume that the forward price equals the futures price.
- 5.26. It is sometimes argued that a forward exchange rate is an unbiased predictor of future exchange rates. Under what circumstances is this so?
- 5.27. Show that the growth rate in an index futures price equals the excess return on the portfolio underlying the index over the risk-free rate. Assume that the risk-free interest rate and the dividend yield are constant.
- 5.28. Explain carefully what is meant by the expected price of a commodity on a particular future date. Suppose that the futures price for crude oil declines with the maturity of the contract at the rate of 2% per year. Assume that speculators tend to be short crude oil futures and hedgers tend to be long. What does the Keynes and Hicks argument imply about the expected future price of oil?
- 5.29. The Value Line Index is designed to reflect changes in the value of a portfolio of over 1,600 equally weighted stocks. Prior to March 9, 1988, the change in the index from one day to the next was calculated as the *geometric* average of the changes in the prices of the stocks underlying the index. In these circumstances, does equation (5.8) correctly relate the futures price of the index to its cash price? If not, does the equation overstate or understate the futures price?
- 5.30. A U.S. company is interested in using the futures contracts traded by the CME Group to hedge its Australian dollar exposure. Define r as the interest rate (all maturities) on the U.S. dollar and r_f as the interest rate (all maturities) on the Australian dollar. Assume that r and r_f are constant and that the company uses a contract expiring at time T to hedge an exposure at time t ($T > t$).
 - (a) Show that the optimal hedge ratio is $e^{(r_f - r)(T - t)}$, ignoring daily settlement.
 - (b) Show that, when t is 1 day, the optimal hedge ratio is almost exactly S_0/F_0 , where S_0 is the current spot price of the currency and F_0 is the current futures price of the currency for the contract maturing at time T .
 - (c) Show that the company can take account of the daily settlement of futures contracts for a hedge that lasts longer than 1 day by adjusting the hedge ratio so that it always equals the spot price of the currency divided by the futures price of the currency.
- 5.31. What is the cost of carry for:
 - (a) a non-dividend-paying stock
 - (b) a stock index
 - (c) a commodity with storage costs
 - (d) a foreign currency.

- 5.32. The spot exchange rate between the Swiss franc and U.S. dollar is 1.0404 (\$ per franc). Interest rates in the United States and Switzerland are 0.25% and 0% per annum, respectively, with continuous compounding. The 3-month forward exchange rate was 1.0300 (\$ per franc). What arbitrage strategy was possible? How does your answer change if the forward exchange rate is 1.0500 (\$ per franc).
- 5.33. An index is 1,200. The three-month risk-free rate is 3% per annum and the dividend yield over the next three months is 1.2% per annum. The six-month risk-free rate is 3.5% per annum and the dividend yield over the next six months is 1% per annum. Estimate the futures price of the index for three-month and six-month contracts. All interest rates and dividend yields are continuously compounded.
- 5.34. Suppose the current USD/euro exchange rate is 1.2000 dollar per euro. The six-month forward exchange rate is 1.1950. The six-month USD interest rate is 1% per annum continuously compounded. Estimate the six-month euro interest rate.
- 5.35. The spot price of oil is \$50 per barrel and the cost of storing a barrel of oil for one year is \$3, payable at the end of the year. The risk-free interest rate is 5% per annum continuously compounded. What is an upper bound for the one-year futures price of oil?
- 5.36. A company that is uncertain about the exact date when it will pay or receive a foreign currency may try to negotiate with its bank a forward contract that specifies a period during which delivery can be made. The company wants to reserve the right to choose the exact delivery date to fit in with its own cash flows. Put yourself in the position of the bank. How would you price the product that the company wants?
- 5.37. A company enters into a forward contract with a bank to sell a foreign currency for K_1 at time T_1 . The exchange rate at time T_1 proves to be $S_1 (> K_1)$. The company asks the bank if it can roll the contract forward until time $T_2 (> T_1)$ rather than settle at time T_1 . The bank agrees to a new delivery price, K_2 . Explain how K_2 should be calculated.